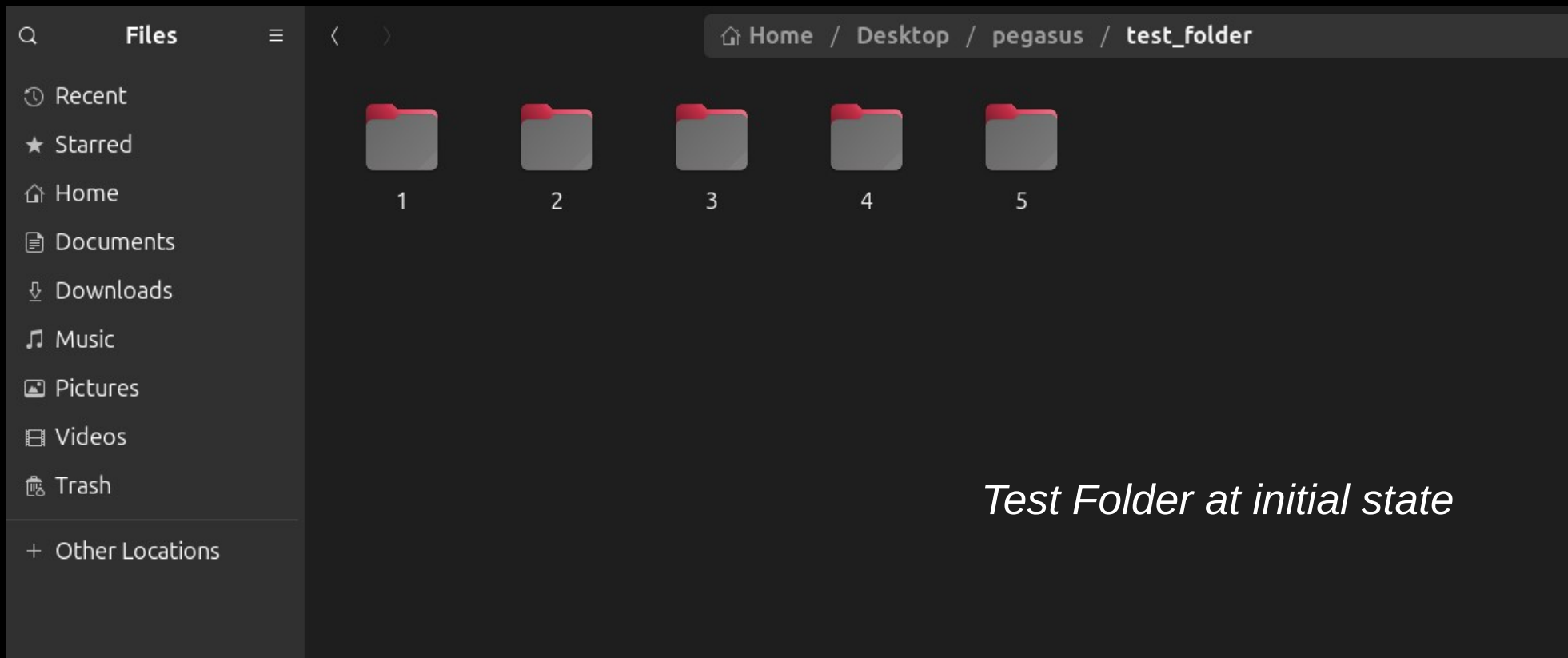


PEGASUS

Visual Overview :

```
(venv) soham-d@krypton:~/Desktop/pegasus$ python -m backend.main  
[BOOT] WELCOME TO PEGASUS....  
[BOOT] AGENT INITIALIZED...  
[INFO] MONITORING ACTIVE...  
█
```

Agent starts running

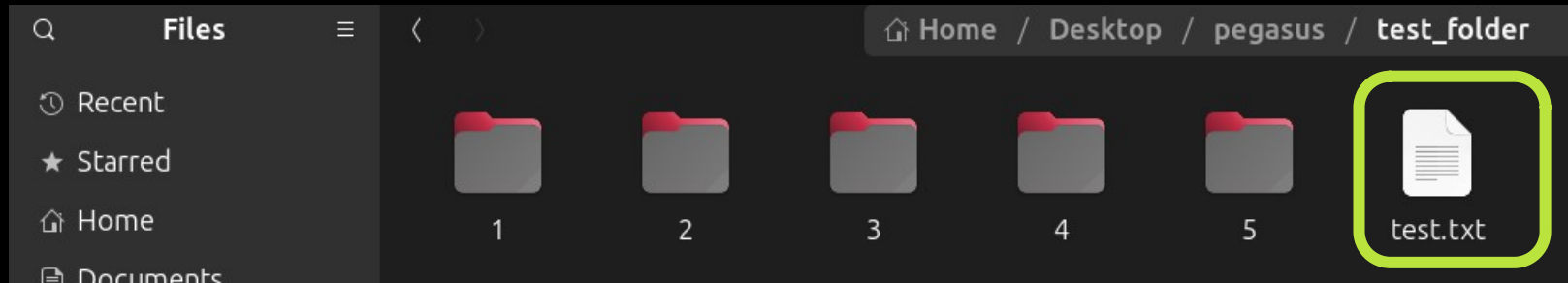


Normal Operations and Agent Behavior :

File Creation :

```
soham-d@krypton:~/Desktop/pegasus/test_folder$ echo "hello pegasus" > test.txt
soham-d@krypton:~/Desktop/pegasus/test_folder$
```

```
(venv) soham-d@krypton:~/Desktop/pegasus$ python -m backend.main
[BOOT] WELCOME TO PEGASUS....
[BOOT] AGENT INITIALIZED...
[INFO] MONITORING ACTIVE
[INFO] OBSERVED FILE SYSTEM EVENT: CREATE...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
```



Open ▾

test.txt

~/Desktop/pegasus/test_folder

hello pegasus

File Modification :

```
soham-d@krypton:~/Desktop/pegasus/test_folder$ echo "edit" >> test.txt
soham-d@krypton:~/Desktop/pegasus/test_folder$
```

```
(venv) soham-d@krypton:~/Desktop/pegasus$ python -m backend.main
[BOOT] WELCOME TO PEGASUS....
[BOOT] AGENT INITIALIZED...
[INFO] MONITORING ACTIVE...
[INFO] OBSERVED FILE SYSTEM EVENT: CREATE...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...

```

Open ▾



test.txt

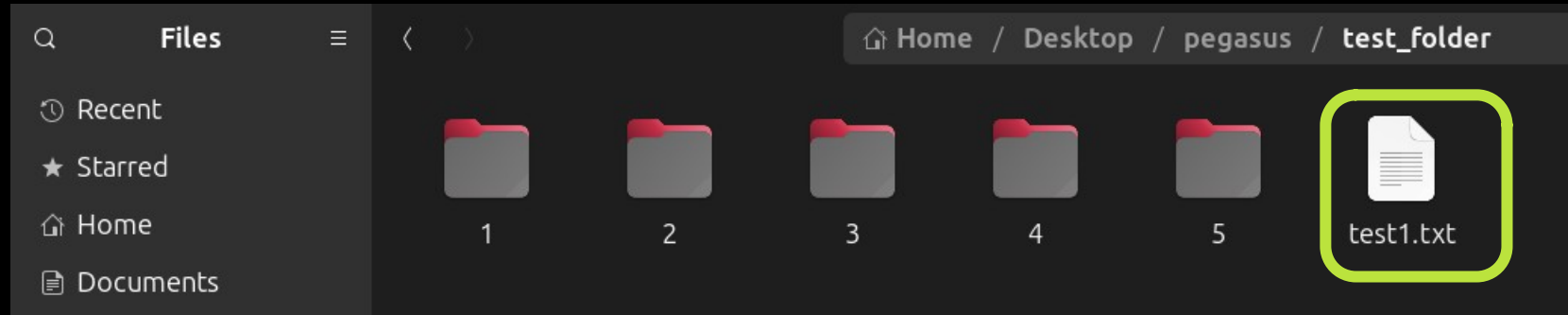
~/Desktop/pegasus/test_folder

```
hello pegasus
edit
```

File Renaming :

```
soham-d@krypton:~/Desktop/pegasus/test_folder$ mv test.txt test1.txt  
soham-d@krypton:~/Desktop/pegasus/test_folder$
```

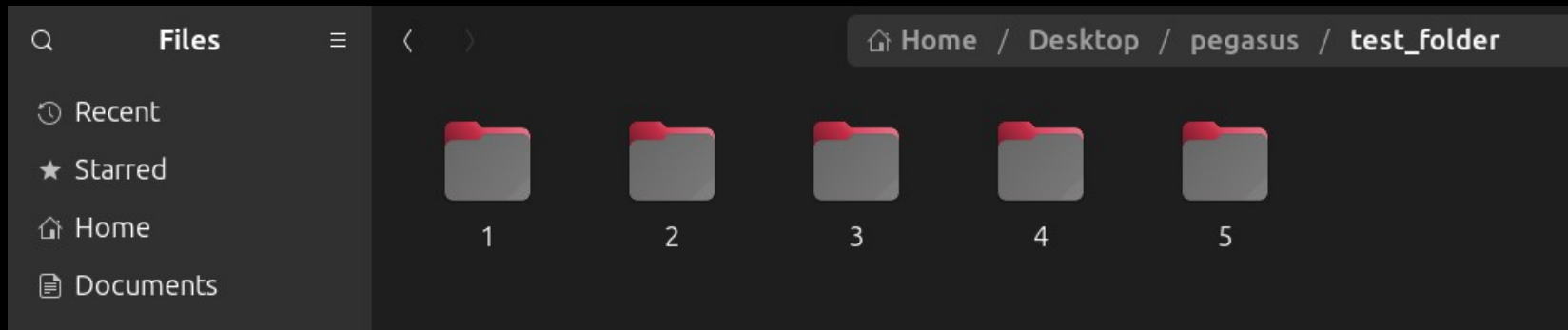
```
(venv) soham-d@krypton:~/Desktop/pegasus$ python -m backend.main  
[BOOT] WELCOME TO PEGASUS....  
[BOOT] AGENT INITIALIZED...  
[INFO] MONITORING ACTIVE...  
[INFO] OBSERVED FILE SYSTEM EVENT: CREATE...  
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...  
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
```



File Deletion :

```
soham-d@krypton:~/Desktop/pegasus/test_folder$ rm test1.txt
soham-d@krypton:~/Desktop/pegasus/test_folder$
```

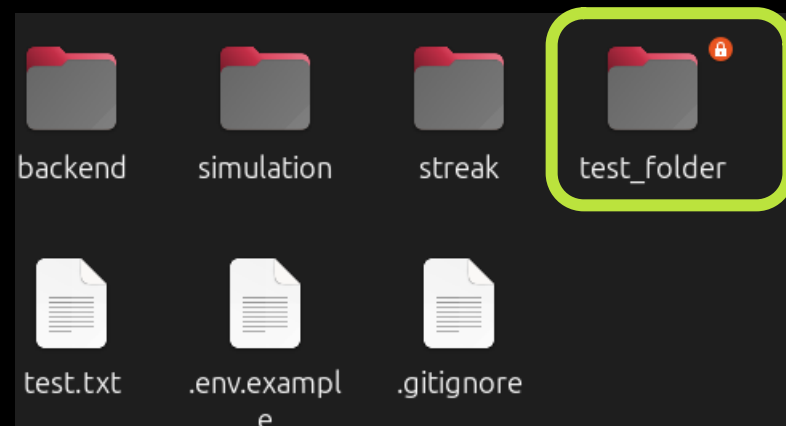
```
(venv) soham-d@krypton:~/Desktop/pegasus$ python -m backend.main
[BOOT] WELCOME TO PEGASUS....
[BOOT] AGENT INITIALIZED...
[INFO] MONITORING ACTIVE...
[INFO] OBSERVED FILE SYSTEM EVENT: CREATE...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
[INFO] OBSERVED FILE SYSTEM EVENT: DELETE...
```



Ransomware Behavior Simulation and Agent Functioning :

Ransomware Simulation :

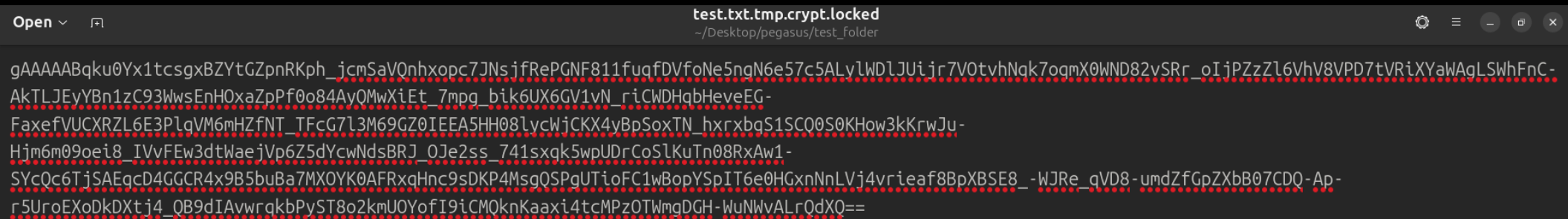
```
soham-d@krypton:~/Desktop/pegasus$ python -m simulation.test_script
[!] Aggressive Ransomware Simulation Started
```

[illegible]

A large number of operations in burst causes the Agent to flag the system behavior as suspicious, thus preventing further operations on the test folder by revoking the modification permissions.

```
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...
```

The rapid modifications keep occurring, and the files start getting overwritten with gibberish (encrypted) data.



Open ▾  test.txt.tmp.crypt.locked
~/Desktop/pegasus/test_folder

gAAAAABqku0Yx1tcsgxBZYtGZpnRKph_jcmSaVQnhxopc7JNsJfRePGNF811fuqfDVfoNe5ngN6e57c5ALylWDLJUijr7V0tvhNqk7oqmX0WND82vSRr_oIjPZzZl6VhV8VPD7tVRiXYaWAgLSWhFnC-
AkTLJEyYBn1zC93WwsEnH0xaZpPf0o84AyOMwXiEt_7mpg_bik6UX6GV1vN_riCWDHqbHeveEG-
FaxefVUCXRZL6E3PlqVM6mHZfNT_TFcG7l3M69GZ0IEEA5HH08lycWjCKX4yBpSoxTN_hxrxbqS1SC00S0KHow3kKrwJu-
Hjm6m09oei8_IVvFEw3dtWaejVp6Z5dYcwNdsBRJ_0Je2ss_741sxqk5wpUDrCoSLKuTn08RxAw1-
SYcQc6TjSAEgcD4GCCR4x9B5buBa7MX0YK0AFRxdHnc9sDKP4Msq0SPgUTioFC1wBopYSpIT6e0HCxnNnLVj4vrieaf8BpXBSE8_-WJRe_qVD8-umdZfGpZXbB07CDQ-Ap-
r5UroEXoDkDXtj4_QB9dIAvwrqkbPyST8o2kmU0YofI9iCMQknKaaxi4tcMPz0TWmgDGH-WuNWvALrQdXQ==

```
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...  
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...  
[INFO] OBSERVED FILE SYSTEM EVENT: MODIFY...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...  
[INFO] OBSERVED FILE SYSTEM EVENT: RENAME...
```

```
SYSTEM IS CRITICAL  
PROCESS KILLED...  
[INFO] RESTORED DATA TO LAST SAFE STATE...
```

As the suspicious behavior continues, the Agent eventually flags the behavior as a serious threat to the system, hence it decides to kill the abnormal process and use the internal real-time snapshots taken till the point, the process has not begun, to restore the corrupted files.

